InsightFlow AI: An Inspectable, Confidence-Aware Pipeline for Conversational Business Intelligence

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***Abstract*—Conversational business intelligence systems translate natural-language questions into database queries, but syntactically valid queries can still fail to preserve the user's intended analytical meaning. This paper presents InsightFlow AI, an inspectable pipeline that combines query planning, confidence-aware response selection, evidence recording, and a semantic correctness layer. The semantic layer checks whether requested measures, filters, grouping, temporal interpretation, join paths, and output structure are preserved through query execution and response generation. The system supports four response actions: answer, warn, clarify, and abstain.**

We report two distinct experiments. In a historical 27-question benchmark on a synthetic sales database, an earlier version of the pipeline executed SQL for all 17 answerable questions and agreed with reference SQL on 15 of those 17 cases. The two disagreements involved diagnostic questions whose intended semantics were not independently adjudicated. In a separate first-run evaluation of the semantic-layer implementation, 66 questions were tested across airline, software-as-a-service (SaaS), and farm database schemas. Of 54 questions labelled answerable, 42 received correct answers, while 11 answerable questions were left unanswered. Two of the 66 cases were classified as silent wrong answers. Both had internal confidence scores near 0.98, indicating that the score alone did not reliably distinguish correct from incorrect outputs. Median and 95th-percentile latency in this run were 14 ms and 38 ms, respectively.

These results expose a practical trade-off between answer coverage and conservative clarification. They also show that explicit semantic checks remain incomplete and that high internal confidence should not be treated as evidence of correctness. The experiments do not establish calibrated confidence, general superiority over existing systems, or elimination of semantic errors. The findings provide an empirical basis for further evaluation of inspectable semantic validation on independently prepared schemas and broader analytical workloads.

***Index Terms*—Conversational business intelligence, text-to-SQL, semantic validation, explainable analytics, confidence-aware systems, selective prediction, query verification**

I. Introduction

Business intelligence systems help organizations examine operational data, monitor performance indicators, and investigate changes in business activity. Traditional interfaces often require users to understand database schemas, query languages, or predefined dashboards. Natural-language interfaces offer an alternative by allowing users to express analytical questions in familiar terms.

However, translating a natural-language question into SQL involves more than generating a query that executes successfully. The generated query must preserve the intended measure, filtering conditions, grouping level, temporal interpretation, relationships between tables, and expected output structure. A query can be syntactically valid and return plausible values while still answering a different question from the one the user asked.

For example, a request to calculate revenue excluding cancelled orders can be misinterpreted as a request to calculate revenue from cancelled orders. A request for yield per acre may produce separate totals for yield and acreage rather than a ratio. A query involving multiple tables may duplicate records if the join relationships or aggregation grain are handled incorrectly. These failures are difficult to detect through syntax validation alone because the database can execute the query without raising an error.

Research on natural-language-to-SQL systems has advanced substantially through datasets and benchmarks such as Spider, BIRD, and Spider 2.0 [2]–[4]. These resources provide increasingly demanding settings for evaluating semantic parsing, database grounding, and query execution. Nevertheless, the success of an individual query execution does not by itself establish that the resulting answer matches the user's intended analytical meaning.

InsightFlow AI is a prototype designed to investigate how conversational business intelligence can combine query planning with inspectable evidence, confidence-aware response selection, and explicit semantic checks. The system records relevant information about the request, the generated SQL, the applied filters, and the resulting data. It also supports

four response actions: ANSWER, WARN, CLARIFY, and ABSTAIN. The intention is to make the system's decision and supporting evidence easier to inspect while allowing unresolved or potentially unsafe requests to be handled conservatively.

A central design consideration is the distinction between confidence and correctness. An internal confidence score can summarize the signals available to a system, but a high score does not establish that a query preserves the user's intended meaning. Accordingly, InsightFlow AI includes a semantic correctness layer that checks selected aspects of the analytical request before and after query execution. The checks cover supported properties such as measure compatibility, filter polarity, grouping, temporal constraints, join paths, and output shape.

This paper evaluates the prototype through two separate experiments. The first is a historical benchmark of an earlier pipeline on a synthetic sales database. The second evaluates the first-run behaviour of the semantic-layer implementation on a sealed set of questions spanning three database schemas. The results are reported separately because the experiments use different system revisions, datasets, and evaluation protocols.

The contributions of this paper are as follows:

1. **Inspectable pipeline design:** We describe a conversational business intelligence pipeline combining query planning, database execution, semantic checks, evidence recording, and confidence-aware response selection.
2. **Two-stage empirical evaluation:** We report historical reference-SQL agreement results and a separate first-run evaluation of semantic validation across airline, SaaS, and farm schemas.
3. **Failure-oriented analysis:** We document examples of incorrect high-confidence answers, unresolved answerable questions, and limitations in the current semantic checks. We distinguish initial results from improvements measured after the failure cases had been inspected.

The paper does not claim that InsightFlow AI outperforms established text-to-SQL systems. Instead, it presents a bounded evaluation of the current prototype, with emphasis on inspectability, semantic failure detection, and the relationship between answer coverage and incorrect responses.

II. Related Work

A. Natural-Language-to-SQL Systems

Seq2SQL explored the generation of structured queries from natural-language questions using reinforcement learning [1]. Spider introduced a large, cross-domain benchmark requiring models to generate SQL for databases that may not have appeared during training [2]. BIRD expanded the evaluation setting through large databases and challenges involving value grounding and real-world database content [3]. Spider 2.0 considers more complex enterprise workflows and database environments [4].

These studies provide important foundations for evaluating natural-language-to-SQL capabilities. InsightFlow AI addresses a complementary system-level concern: after a candidate query has been constructed, how can the system inspect whether the query preserves the intended analytical request?

The present evaluation is not a direct comparison with these benchmarks. It uses project-specific synthetic datasets and does not establish comparative performance against the systems evaluated in the cited work.

B. Selective Prediction and Abstention

Selective prediction studies systems that may decline to make a prediction when the available evidence is insufficient. This creates a trade-off between coverage, which measures how often a system responds, and the risk of error among the responses it provides [5].

The same trade-off arises in conversational business intelligence. A system that asks for clarification on every difficult request may avoid some incorrect answers but provide limited analytical coverage. A system that answers aggressively may be more responsive while producing plausible but incorrect results.

InsightFlow AI implements response actions intended to distinguish direct answers from warnings, clarification requests, and abstentions. Its evaluation therefore considers both correctness and the handling of questions that should not be answered directly.

C. Confidence Estimation and Calibration

Confidence estimates can support decisions about whether to answer or request clarification. However, confidence is useful only when its relationship with observed correctness is understood.

Expected calibration error (ECE) is one commonly used summary of calibration. It compares average confidence with empirical accuracy across confidence bins [6]. Its interpretation depends on the population, binning strategy, and definition of correctness.

The earlier InsightFlow AI pipeline combines several internal signals into a composite confidence score. The newer semantic-layer evaluation examines whether high scores correspond to correct outputs. The results show why confidence, correctness, coverage, and silent error counts must be reported separately rather than treating a confidence value as a guarantee.

D. Evidence and Tool-Oriented Systems

ReAct combines reasoning and action in language-model-based systems [7]. The Model Context Protocol (MCP) defines a standard interface through which applications can expose tools and contextual information to clients [8].

InsightFlow AI includes evidence-recording capabilities and tool-server integration in its broader implementation. However, the experiments reported in this paper do not evaluate MCP interoperability or compare tool-use frameworks. Evidence visibility is treated as an inspectability feature, not as proof that the generated query is correct.

TABLE I
RELATED WORK AND ITS ROLE IN THIS PAPER

Work	Contribution and focus	Role in this paper
Seq2SQL [1]	Method: natural-language-to-SQL generation with reinforcement learning	Background; not compared
Spider [2]	Benchmark: cross-domain text-to-SQL	Background; not evaluated on
BIRD [3]	Benchmark: large-scale, database-grounded text-to-SQL with external knowledge	Background; not evaluated on
Spider 2.0 [4]	Benchmark: enterprise text-to-SQL workflows	Background; not evaluated on
Selective classification [5]	Method: abstention under a risk–coverage trade-off	Motivates the answer/clarify trade-off
Bayesian binning [6]	Method: calibration; expected calibration error	ECE used as a diagnostic
ReAct [7]	Method: interleaved reasoning and tool actions	Context; not evaluated
MCP [8]	Protocol: exposing tools and context to applications	Servers implemented; not evaluated
InsightFlow AI (this work)	Prototype: typed plan, semantic checks, four actions, evidence record	Evaluated on synthetic SQLite schemas only

Comparison of scope and role only; no system was compared experimentally.

Table I summarizes the cited work and its role in this paper.

III. SYSTEM DESIGN

A. Architecture Overview

InsightFlow AI is implemented as a web-based conversational business intelligence prototype. The frontend uses Next.js 14.2.5, while the backend uses Python FastAPI and SQLAlchemy for database access. SQLite is the default database used in the reported experiments. PostgreSQL connectivity has not been evaluated as part of the reported results.

The default planner is deterministic and rule-based. An optional language-model client exists in the broader implementation, but it was disabled in the reported runs. Consequently, the results should not be interpreted as an evaluation of an LLM-driven query planner.

The pipeline can be summarized in the following stages:

1. Receive the natural-language question and available database context.
2. Identify the analytical intent and construct a candidate plan.
3. Resolve relevant schema elements and candidate measures.
4. Construct a candidate SQL query.
5. Apply query and execution-safety checks.
6. Execute the query against the selected database.
7. Check selected semantic properties against the request and result.
8. Assemble the supporting evidence record.
9. Select a response action and present the result, warning, clarification request, or abstention.

The stages provide inspection points for diagnosing failures. The pipeline does not assume that successful execution is sufficient to establish correctness.

Fig. 1 shows these components and the data flow. The evaluation harness calls the orchestrator directly, so the web frontend, the REST API and the MCP tool servers were not exercised by the reported experiments.

B. Response Actions

The system supports four response actions.

ANSWER: The system returns a result when the applicable checks do not identify a blocking issue.

WARN: The system returns a result with a caveat or stated assumption when an issue has been identified but does not trigger a complete block.

CLARIFY: The system requests additional information when the analytical intent or necessary schema interpretation cannot be resolved sufficiently.

ABSTAIN: The system declines to produce a direct answer when the request falls outside supported capabilities or cannot be handled within the current constraints.

These actions are intended to make uncertainty operationally visible. They do not imply that every incorrect query will be detected or that every request requiring clarification will be identified correctly.

C. Confidence-Aware Decision Signal

The earlier pipeline computes a composite confidence score from nine signals:

- SQL validity;
- schema match;
- KPI match;
- context consistency;
- data completeness;
- evidence strength;
- result consistency;
- intent coverage; and
- plan fidelity.

The associated weights are 0.12, 0.10, 0.13, 0.13, 0.08, 0.07, 0.10, 0.13, and 0.14, respectively. These weights sum to 1.00.

The historical implementation used a high threshold of 0.70 and a low threshold of 0.40. It also capped the composite score using intent coverage and plan fidelity:

$$C = \min\big(S,\ \text{coverage} + 0.05,\ \text{fidelity} + 0.05\big), \quad (1)$$

where S is the weighted signal score.

The score is an internal decision signal, not a statistically calibrated probability that an answer is correct. The first-run semantic-layer evaluation includes high-confidence incorrect outputs, which demonstrates the importance of evaluating confidence against observed outcomes.

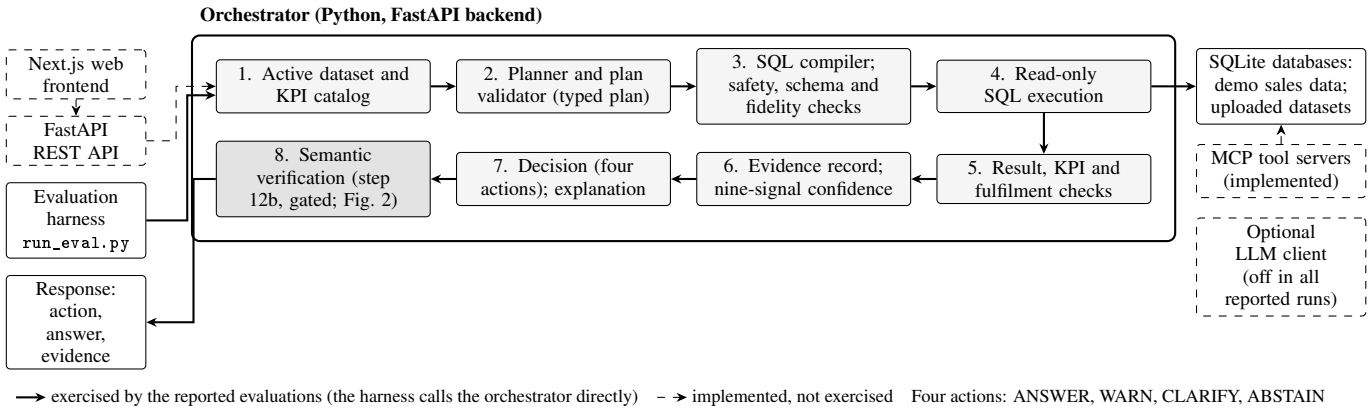


Fig. 1. System architecture of InsightFlow AI, derived from the orchestrator pipeline; the stages shown are present at revisions `78d906b` (K1), `a63cec1` (K3) and `b11793b`. Solid arrows mark paths exercised by the reported evaluations, whose harness calls the orchestrator directly; dashed elements are implemented but were not exercised. The semantic verification stage applies only under the conditions shown in Fig. 2.

D. Semantic Correctness Layer

The newer implementation adds a semantic validation package containing a semantic specification, type checks that compare the specification with the wording of the question, preservation checks against the executed SQL, the result rows and the written answer, and a verifier that returns findings classified as violations or notes. The package runs after execution, as step 12b of the pipeline (Fig. 2); the type checks do not use the executed SQL or rows.

The layer checks selected properties of a request and its execution plan, including:

1. **Measure compatibility:** Whether the requested measure is compatible with the type and meaning of the referenced column.
2. **Filter preservation:** Whether exclusions, negation, and filter polarity are preserved in the generated query.
3. **Aggregation and output shape:** Whether the query produces the expected scalar or grouped result structure.
4. **Temporal interpretation:** Whether date constraints and the selected date column are consistent with the request.
5. **Join grounding:** Whether the join path is supported by available schema relationships and whether the join may multiply records.
6. **Ratio handling:** Whether a request for a ratio is represented by a ratio calculation rather than separate totals. This check was added after the K1 run, in response to case V07; it is part of the K3 configuration but was not present at revision `78d906b`.
7. **Result and explanation consistency:** Whether numerical claims in the explanation correspond to the query result and whether the result structure matches the request.

A detected violation changes a pending ANSWER or WARN to CLARIFY and caps the confidence at 0.30. Findings of severity note are recorded in the response diagnostics and do not change the action. The layer runs only when the pending action is ANSWER or WARN, the dataset is not the built-in demo dataset, SQL was executed successfully, and the plan has one of the intent kinds direct KPI, breakdown, top-N or trend; the planner's `ratio` intent kind ("X per Y over KPIs) was added in revision `a63cec1`. Other intents and the demo dataset are not checked by this layer. Table II lists the implemented rules. If the verifier itself raised an exception, revisions `78d906b` and `a63cec1` logged the error and left the decision unchanged (fail-open); revision `b11793b`, committed after both Round 6 runs, returns CLARIFY instead (fail-closed). No Round 6 results are reported for that revision.

The semantic layer is rule-based and only covers selected classes of errors. It does not formally prove equivalence between the user's intended question and the generated SQL. Unrecognized wording, incomplete schema information, unsupported analytical patterns, or limitations in the rules may still result in an incorrect response.

E. Evidence Record

The evidence record is intended to expose the information needed to inspect how an answer was produced. Depending on the execution path, it can include the original question, generated SQL, KPI definition, applied filters, calculations, result row count, and selected result rows.

This record can help distinguish failures in query execution from failures in interpreting the question. It also supports reproducibility and debugging by making the intermediate query and its result available for examination.

Evidence completeness and evidence correctness are separate properties. A record may contain a valid SQL statement and the values returned by the database while still documenting a query that applies the wrong filter or aggregation.

F. Tool Integration

The broader project includes tool-server integration intended to expose selected data and analysis operations through structured interfaces. The implementation inventory records multiple tool groups and server components. However, the experiments in this paper do not measure the reliability,

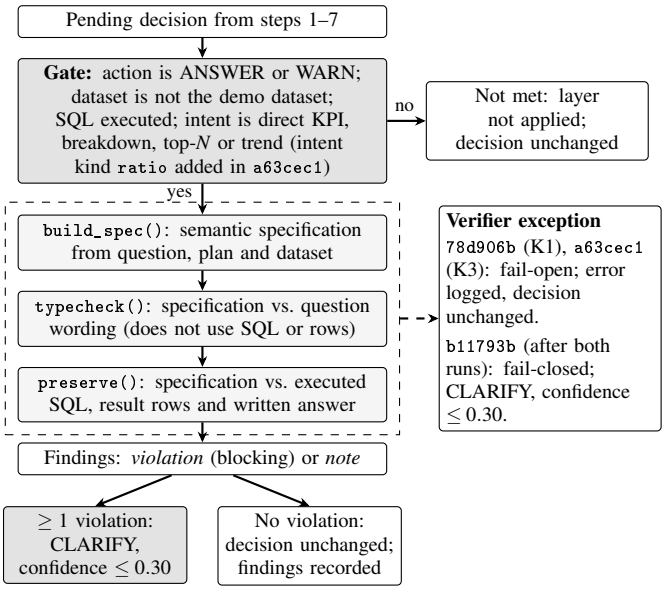


Fig. 2. Semantic-layer workflow (step 12b). The gate restricts verification to non-demo datasets, executed SQL and the listed intent kinds. A violation turns a pending answer into CLARIFY with confidence at most 0.30. Verifier exceptions were handled fail-open at the K1 and K3 revisions and fail-closed from `b11793b`, after both runs.

interoperability, or performance of those integrations. They are therefore not treated as experimentally validated contributions in the reported results.

IV. Evaluation Methodology

The evaluation comprises two experiments with different purposes. The historical benchmark measures an earlier pipeline on a synthetic sales database. The Round 6 evaluation measures the first-run behaviour of the newer semantic-layer implementation across three schemas. Results from these experiments are not pooled.

A. Historical Sales Benchmark

The historical benchmark uses a synthetic sales database containing:

- 2,037 orders;
- 60 customers;
- 12 products;
- four product categories;
- four regions; and
- records covering January through September 2026.

The synthetic data include a revenue decline in July for one region/category combination. The benchmark contains 27 author-written questions, classified as 17 answerable, four ambiguous, and six out of scope.

Five configurations were evaluated, producing 135 recorded runs in total. The historical results are tied to commit `3c843de88ca0b62669749b217aaef000e3e6363e`. They are not measurements of the later semantic-layer implementation.

TABLE II
Implemented Semantic Rules at Revision a63cec1

Rule codes	Detects	Since
Type checks: specification vs. question wording		
`unresolved_reference`	Unresolved part of the question when an answer was produced	K1
`flag_used_as_quantity`, `identifier_used_as_quantity`	Aggregating a yes/no flag or an identifier as an amount	K1
`entity_count_mismatch`	"How many X" answered by counting rows of another table	K1
`shape_scalar_expected`, `shape_breakdown_expected`, `shape_ranking_expected`	Plan shape differs from the wording (single figure, breakdown, ranked list)	K1
`date_role_guessed`, `date_role_mismatch`	Date column chosen without basis, or differing from the one named	K1
`unsafe_join`, `unsafe_grouping_path`	Join or grouping through a relationship that is not safely many-to-one	K1
`polarity_inverted`	Wording excludes a value that the query keeps	K3
`ratio_unsupported`	"Per X" ratio of two measures that the query does not compute	K3
`assumption` (note)	Recorded assumption; does not block	K1
Preservation checks: executed SQL, rows and written answer		
`measure_not_preserved`	Required aggregate missing from the SQL	K1
`filter_not_preserved`, `exclusion_not_preserved`	Planned filter missing, or an exclusion lost its negation	K1
`time_not_preserved`	Planned time bound missing	K1
`grouping_not_preserved`, `ranking_not_preserved`	Dimension missing from GROUP BY; top-N not applied	K1
`unplanned_join`, `fanout_join`	Unneeded join, or one-to-many join that multiplies rows	K1
`unexpected_columns`, `shape_violation`	Result columns or row count differ from the expected shape	K1
`answer_not_faithful`	Written answer states a number absent from the result	K1

All rules are blocking violations except `assumption`. K1: present at `78d906b`; K3: added in `a63cec1`, after the K1 run. Rules apply only under the gate conditions in Fig. 2.

The principal measures were SQL execution success, reference-SQL agreement, routing for non-answerable questions, question coverage, latency, and confidence calibration.

Reference-SQL agreement measures whether the generated query result agrees with the designated reference query under the benchmark's comparison procedure. It is not necessarily equivalent to independently adjudicated semantic correctness. If a reference query does not fully specify the intended analytical comparison, agreement or disagreement alone may be insufficient to determine whether the generated answer is semantically correct.

B. Round 6 Semantic-Layer Evaluation

The Round 6 evaluation contains 66 questions across three database schemas:

- airline data involving flights;
- SaaS data involving invoices; and
- farm data involving readings and harvests.

Reference results were prepared using independent SQLite SQL. The question set was sealed before the first run, and the implementation was frozen at commit `78d906b` before the sealed questions were inspected. This first-run configuration is identified as K1.

The evaluation includes answerable questions and questions expected to require clarification or otherwise not produce a direct answer. The recorded measures include:

- correct answers among answerable questions;
- answered questions among answerable questions;
- answerable questions left unanswered;
- silent wrong answers among all questions;
- correct clarification decisions; and
- median and 95th-percentile latency.

A silent wrong answer is a returned answer classified as incorrect by the recorded evaluation reference, including cases where the expected outcome was clarification rather than a direct answer.

These measures have different denominators. In particular, the number of answerable questions answered should not be confused with the total number of answer outputs across all question classes. Results should be interpreted using the denominator specified for each metric.

C. Post-Hoc Tuning Protocol

After K1 was recorded, failure cases were inspected and implementation changes were made. The resulting K3 configuration was evaluated on the same set. Since the failure cases had already informed development, K3 is a post-hoc tuned result, not an independent held-out result.

K3 is included to document the observed effect of targeted fixes. K1 remains the primary first-run result for the sealed set. Neither result alone establishes generalization to unseen production databases.

D. Reproducibility

The project source code and evaluation artifacts are maintained in the repository [9]:

https://github.com/jagan1344/insightflow-ai

The Round 6 evaluation includes a seal file containing SHA-256 digests for the question and fixture files. The first-run implementation revision and historical benchmark revision are recorded separately so that the results can be interpreted in relation to the corresponding code.

The reported latency values are local experimental measurements. They should not be treated as service-level guarantees because the evaluation does not establish production performance under network latency, concurrent requests, or deployment-level load.

V. Results

A. Historical Benchmark Results

In the historical 27-question benchmark, the earlier pipeline emitted and executed SQL for all 17 answerable questions. The generated results agreed with reference SQL on 15 of the 17 answerable cases, corresponding to 88.2% reference-SQL agreement.

All 10 non-answerable questions were routed to clarification without SQL execution. Overall coverage was 17 out of 27 questions. The reported latency was 4 ms at the median and 6 ms at the 95th percentile.

The historical composite ECE was 0.416. An answer-only ECE calculation was 0.0978. These values refer to different evaluation populations and should not be interpreted as interchangeable estimates of calibration.

The two reference-SQL disagreements occurred on diagnostic questions asking why a metric had changed. The designated reference queries did not fully establish the intended comparison, and the disagreements were not independently adjudicated. We therefore report the result as 15/17 reference-SQL agreement rather than as a validated estimate of semantic accuracy.

The five historical configurations produced identical outcomes on this benchmark. Thus, the ablation experiment did not demonstrate a measurable difference among those configurations under the tested questions. This result does not establish that the removed components would be ineffective on other datasets or workloads.

B. Round 6 K1 First-Run Results

The Round 6 K1 evaluation tested 66 questions across three schemas. Of the 54 questions labelled answerable, 42 received correct answers. The system answered 43 of the 54 answerable questions and left 11 unanswered. Two of the 66 total cases were classified as silent wrong answers. Correct clarification was recorded for 11 of the 12 cases labelled as requiring clarification.

Median latency was 14 ms, and the 95th-percentile latency was 38 ms. Table III lists these measures alongside the historical benchmark and K3, and Fig. 4 compares K1 with K3.

The difference between the answerable-question measures is important. The count of 42 correct answers describes correctness against the recorded reference among answerable questions. The count of 43 answered questions describes coverage, not correctness. We therefore report them separately.

C. Analysis of Silent Wrong Answers

Two K1 cases were classified as silent wrong answers. They demonstrate distinct semantic failure modes.

Case A17: Reversed exclusion filter. The question asked for total fare excluding cancelled flights. The generated SQL used the condition `is_cancelled = 1`, selecting cancelled flights rather than excluding them. The result was 45,804.48, compared with the reference value of 618,324.80. The system returned this answer with an internal confidence score of approximately 0.979.

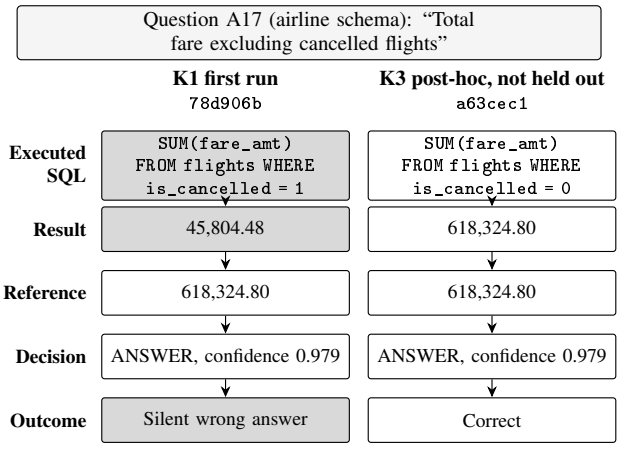
Fig. 3 shows the recorded traces for A17 in the K1 and K3 runs.

This case illustrates that a query can be valid and executable while reversing the meaning of a negated condition. Syntax checks alone would not identify the error.

TABLE III
HISTORICAL AND ROUND 6 RESULTS: DISTINCT REVISIONS, QUESTION SETS AND DENOMINATORS

	Historical	K1	K3
Revision	`3c843de`	`78d906b`	`a63cec1`
Run type	First recorded run	First run, sealed set	Post-hoc, not held out
Questions (answerable / other)	27 (17 / 10)	66 (54 / 12)	66 (54 / 12)
Correct / answerable	15/17[a]	42/54	49/54
Answered / answerable	17/17	43/54	49/54
Unanswered / answerable	0/17	11/54	5/54
Silent wrong / all questions	n/a[b]	2/66	0/66 observed
Correct clarifications / labelled	10/10	11/12	12/12
Mean confidence, correct answers	—	0.976	0.977
Mean confidence, silent wrong	—	0.979	—
Latency, median / p95 (ms)	4 / 6	14 / 38	15 / 43

[a] Agreement with the reference SQL, not adjudicated semantic correctness. [b] Not measured on the same basis: two answered questions disagreed with the reference SQL and were not adjudicated. K3 was obtained after inspecting the K1 failures and is not held-out evidence.



Shaded cells differ from the reference behaviour. Values and actions are copied from the committed run records; SQL shown without identifier quotes.

Fig. 3. Recorded execution trace for question A17 in the K1 first run (`78d906b`) and the post-hoc K3 run (`a63cec1`). SQL, values, decisions and outcomes are copied from the committed run records; K3 is not held out.

Case V07: Unsupported ratio request. The question asked for total yield in tons per acre. The system returned separate sums for yield and acreage rather than calculating the requested ratio. The evaluation reference indicated that the system should have requested clarification. The answer was returned with a WARN action and an internal confidence score of approximately 0.979.

This case illustrates the difference between returning relevant measures and computing the requested analytical quantity. The presence of both yield and acreage in the result does not establish that the requested ratio has been calculated.

Both cases carried high internal confidence scores. In this evaluation, confidence alone did not reliably distinguish these incorrect outputs from correct answers.

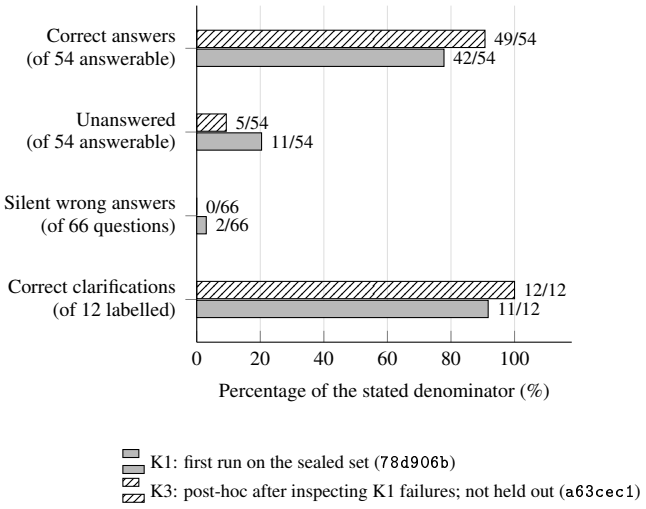


Fig. 4. Round 6 K1 first run (`78d906b`) versus K3 post-hoc run (`a63cec1`) on the same 66 sealed questions. Each bar is a percentage of the denominator stated on its label; counts are printed beside the bars. K3 was measured after the K1 failures had been inspected and is not held out.

D. Unanswered Answerable Questions

The K1 system left 11 answerable questions unanswered. Recorded failure patterns included unfamiliar foreign-key names, abbreviations, flags represented in related tables, negation phrasing, word-stem variants, and numeric comparisons.

Table IV summarizes the observed failures, their status after the post-hoc changes, and the remaining limitations.

These cases indicate that conservative handling can limit coverage even when a query may be answerable with the available data. The result highlights two separate objectives for future work: preventing unsupported answers and reducing unnecessary clarification.

A useful semantic validation system must address both. Excessive clarification can make a system less useful, while overly permissive query generation can produce plausible but incorrect results. Reporting only the number of correct answers would not adequately describe this trade-off.

E. Post-Hoc K3 Results

After the K1 failure cases had been inspected, the tuned K3 configuration achieved the following results on the same 66-question set:

- 49 correct answers among 54 answerable questions;
- 49 answered questions among 54 answerable questions;
- zero observed silent wrong answers among the 66 questions;
- 12 correct clarifications among 12 clarification-labelled cases; and
- median and 95th-percentile latency of 15 ms and 43 ms.

Table III and Fig. 4 compare the K1 and K3 results.

TABLE IV
ERROR ANALYSIS AND LIMITATIONS

Case or class	Observed	After post-hoc changes (K3, not held out)	Remaining limitation
A17: reversed exclusion	K1: `is_cancelled = 1`; 45,804.48 vs. reference 618,324.80; ANSWER at 0.979; silent wrong	Correct (`is_cancelled = 0`); rule `polarity_inverted` added	Fix informed by the K1 failure; not tested on held-out questions
V07: "per acre" ratio	K1: two separate sums; WARN at 0.979; silent wrong	CLARIFY at 0.30; rule `ratio_unsupported` added	Ratios of two measures are clarified, not computed
Unanswered answerable questions	K1: 11/54; question families: numeric filter 2, abbreviation 2; undetected foreign key, time, nulls, related flag, fan-out, negation, flag 1 each	5/54; families: abbreviation, flag, numeric filter, single value, undetected foreign key 1 each	Coverage is traded for conservative clarification
Historical D01, D02	`3c843de`: reference SQL returns only the July total; disagreements not adjudicated	Not applicable	Requires semantic reference annotation
Verifier exceptions	Fail-open at `78d906b` and `a63cec1`	Fail-open	Fail-closed from `b11793b`; Round 6 not re-run at that revision
Scope of the semantic layer	Not applied to the demo dataset or to intents other than direct KPI, breakdown, top-N, trend	`ratio` intent kind added	Other intents rely on the earlier checks only
Unsupported patterns	Ratios of measures, period-over-period with dimensions, share of total over joins, many-to-many grouping, anti-joins, non-ISO dates, undeclared foreign keys, non-SQLite engines, non-English questions	Unchanged	Outside the current rules

K3 reflects development informed by observed failures. The result is evidence that targeted changes addressed the evaluated cases, but it is not an independent estimate of performance on unseen data. K1 therefore remains the primary first-run result, and K3 is not evidence that the semantic layer eliminates silent errors.

F. Confidence Across First-Run Evaluation Sets

The Round 6 calibration report also aggregates results from multiple first-run evaluation sets. In the pooled report, 23 silent wrong answers were recorded among 177 outputs with confidence at least 0.9. This aggregate spans multiple question sets and is not the K1-only rate.

The pooled result reinforces the K1 observation that high confidence does not guarantee correctness. Its scope is pooled across sets, and it is not a substitute for a K1-specific calibration analysis.

A complete calibration study would require an explicitly defined evaluation population, stable semantic labels, documented treatment of clarification and abstention, and a consistent calculation procedure. The current evidence is sufficient to motivate that analysis but does not establish that the confidence score is calibrated.

VI. DISCUSSION

The experiments provide several observations relevant to the design of conversational business intelligence systems.

First, successful SQL execution is not sufficient evidence that an analytical request has been answered correctly. The historical benchmark demonstrated high execution success, but two reference-query disagreements remained unresolved because the intended diagnostic comparison was not fully specified. The K1 semantic-layer evaluation provided more concrete examples: a reversed exclusion filter and an unsupported ratio request both resulted in incorrect high-confidence answers.

Second, semantic checks need to examine the meaning of analytical operations, not only the validity of the SQL statement or the existence of referenced columns. Negation, ratios, temporal interpretation, join relationships, and aggregation grain are all possible sources of semantic mismatch. The current checks cover selected examples of these issues, but the K1 errors show that their coverage is incomplete.

Third, confidence should be treated as one input to response selection rather than as a substitute for semantic verification. The two K1 silent wrong answers had mean confidence close to 0.98, comparable to the confidence recorded for correct answers. This observation does not mean that confidence estimates are useless in every setting. It means that their reliability must be established empirically and that high scores should not override explicit semantic violations.

Fourth, answer coverage and error risk must be evaluated jointly. K1 left 11 answerable questions unanswered, including questions affected by abbreviations and schema relations. These cases show that conservative handling can avoid some unsupported responses while also preventing valid questions from being answered. Future evaluation should report coverage, correctness, silent wrong answers, and clarification quality together.

Finally, the difference between K1 and K3 illustrates the importance of separating first-run and post-hoc results. Targeted fixes can improve performance on known failure cases, but those improvements do not provide an unbiased estimate of generalization. A stronger evaluation should freeze the implementation before independent question sets are revealed and should reserve an untouched set for final testing.

VII. Limitations

This study has several limitations.

Synthetic datasets and limited scale. The experiments use relatively small, synthetic datasets and author-written questions. The results do not establish performance on production enterprise data, broader user populations, or naturally occurring business questions.

Reference ambiguity. Reference SQL does not always fully specify the intended analytical meaning of a question. This limitation is particularly relevant to diagnostic questions involving explanations for changes in metrics. Future evaluation should include semantic reference annotations and independent adjudication of ambiguous cases.

Database coverage. All reported database experiments use SQLite. Performance and correctness on PostgreSQL or other database engines have not been established by these experiments.

Partial semantic coverage. The semantic layer is rule-based and does not cover every possible query pattern. Known limitations include ratios of measures, period-over-period comparisons combined with dimensions, share-of-total calculations over joins, many-to-many grouping, anti-joins, non-ISO date formats, and foreign-key relationships that are not declared or inferable. Heuristic synonyms and word stems may also over-trigger or fail to identify intended concepts. The layer is not applied to the built-in demo dataset or to intents other than direct KPI, breakdown, top-N, trend and, from revision `a63cec1`, the `ratio` intent kind (Fig. 2).

Post-hoc evaluation. K3 was tuned after K1 failure cases had been inspected. The K3 results are therefore not independent held-out evidence and should not be used as the main estimate of generalization.

Confidence calibration. The observed high-confidence errors show that the current confidence score is not a sufficient guarantee of correctness. The reported values do not establish calibrated probabilities, and the pooled calibration report covers several different evaluation sets.

Latency and deployment. Latency measurements come from local runs. They do not establish performance under production networking, concurrent workloads, or deployment-level conditions.

External comparison. The study does not provide a controlled comparison against established text-to-SQL baselines on the same question set, schemas, and execution protocol. The results therefore do not support a claim of superiority over existing systems.

VIII. Conclusion and Future Work

This paper presented InsightFlow AI, an inspectable conversational business intelligence prototype combining query planning, confidence-aware response selection, evidence recording, and a rule-based semantic correctness layer.

The historical sales benchmark showed that the earlier pipeline executed SQL for all 17 answerable questions and agreed with reference SQL on 15 of those cases. However, the two disagreements were not independently adjudicated for intended semantics, and the five historical configurations produced identical outcomes on the benchmark.

In the separate Round 6 first-run evaluation, the semantic-layer implementation produced 42 correct answers among 54 answerable questions, left 11 answerable questions unanswered, and generated two silent wrong answers among 66 questions. Both silent errors carried internal confidence scores near 0.98. These findings show that semantic errors remain possible even when a system exposes confidence and performs explicit checks.

The post-hoc K3 results improved on the evaluated cases, but they should be interpreted as targeted engineering results rather than independent evidence of generalization. The overall findings support further investigation of inspectable semantic validation, while underscoring the need to evaluate correctness, coverage, clarification behaviour, and confidence together.

Future work should focus on independently authored and sealed question sets, semantic reference annotations reviewed by independent evaluators, broader schema and database coverage, and controlled comparisons with suitable baseline systems. Evaluation should also examine whether the semantic layer improves the balance between useful answer coverage and the risk of silent wrong answers on unseen data.

The current results do not establish that semantic errors have been eliminated or that InsightFlow AI is superior to existing text-to-SQL systems. They provide a reproducible starting point for examining how explicit semantic checks can support more inspectable conversational analytics.

Reproducibility Statement

The project source code and evaluation artifacts are maintained at:

https://github.com/jagan1344/insightflow-ai

The historical sales benchmark is associated with commit `3c843de88ca0b62669749b217aaef000e3e6363e`. The Round 6 first-run implementation was frozen at commit `78d906b` before the sealed questions were inspected. The K3 results were obtained after inspection and tuning and are reported separately.

The Round 6 seal file, `evaluation/v3/FINAL6_SEAL.txt`, records SHA-256 digests for the sealed question and fixture files. Revision `b11793b`, committed after both Round 6 runs, makes the semantic verifier fail closed on internal errors; at that revision the backend regression suite passed (495 tests, 3 warnings). These tests check implementation behavior; they are not an accuracy benchmark, and no Round 6 results are reported for that revision.

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